PROJECT PROPOSAL

Project Title : ONLINE HOTEL MANAGEMENT SYSTEM

Detail	Information
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Submission Date	03-11-2026

Supervisor :

Introduction

Brief Project Overview

- This project focuses on creating a system that helps hotels manage their daily tasks automatically.
- It includes features for room booking, guest information, billing, and check-in/check-out management.
- The system is designed to make hotel operations faster, more accurate, and easier for staff to handle.

Importance of the Topic

- A Hotel Management System is important because it helps reduce errors and saves time for hotel staff.
- It ensures proper record-keeping, avoids double bookings, and improves service quality for guests.
- In a competitive hospitality industry, such systems help hotels run professionally and make better business decisions.

Problem Statement

Explain the Core Problem

- Manual booking processes cause errors such as double reservations.
- Managing guest records on paper is time-consuming and unsafe.
- Billing and payment calculations often become inaccurate without automation.
- Staff struggle to track room availability in real time.
- Overall hotel operations slow down due to lack of an organized system.

Target Users

- Hotel managers who oversee daily operations.
- Receptionists responsible for bookings and guest check-in/check-out.
- Accounting staff who handle billing and financial records.
- Hotel owners who need reports for business decisions.
- Guests who want fast, smooth, and reliable service.

Objectives

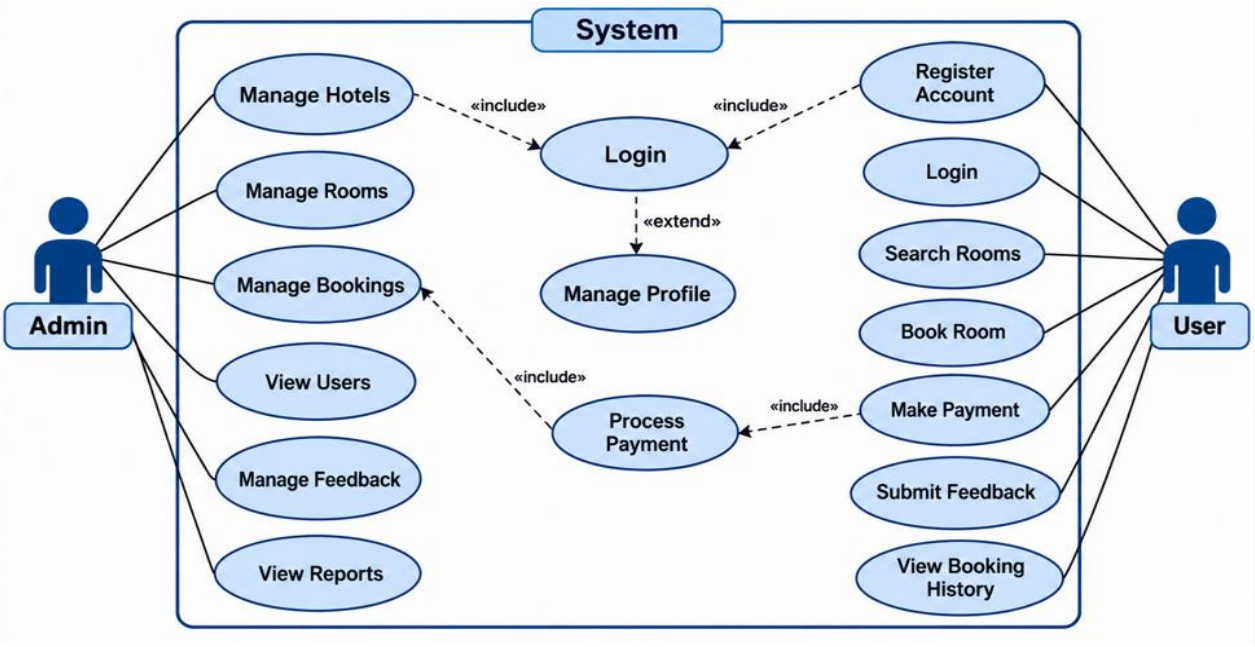
Main Goals of the Project

- To automate hotel operations such as booking, check-in, check-out, and billing.
- To reduce manual errors by storing and managing data digitally.
- To improve service speed and overall guest satisfaction.
- To provide hotel staff with an easy-to-use and efficient management interface.
- To maintain accurate records for better decision-making and reporting.

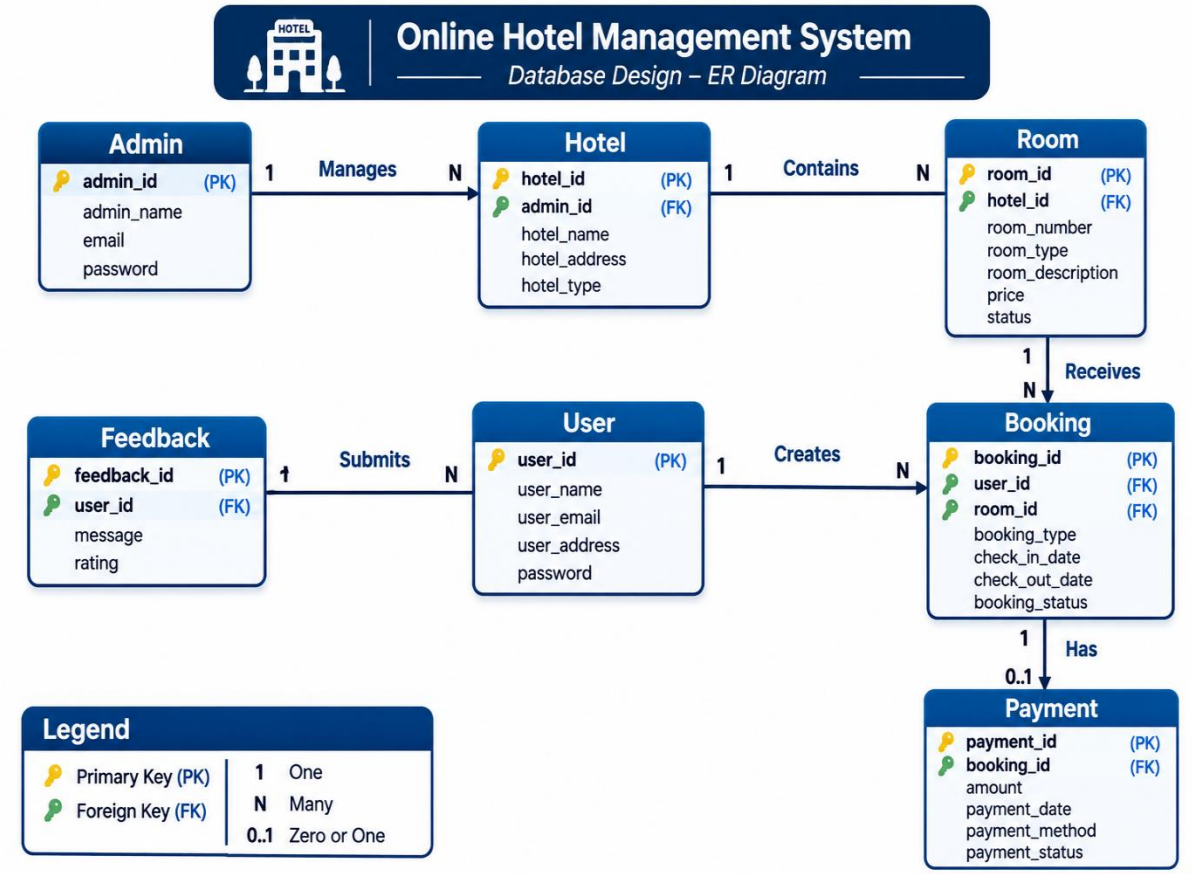
Expected Outcomes

- A fully functional system that simplifies daily hotel tasks.
- Faster and more accurate room reservation and billing processes.
- Improved data security and organized storage of guest information.
- Reduced workload for staff and improved operational efficiency.
- Better customer experience through quick service and smooth management.

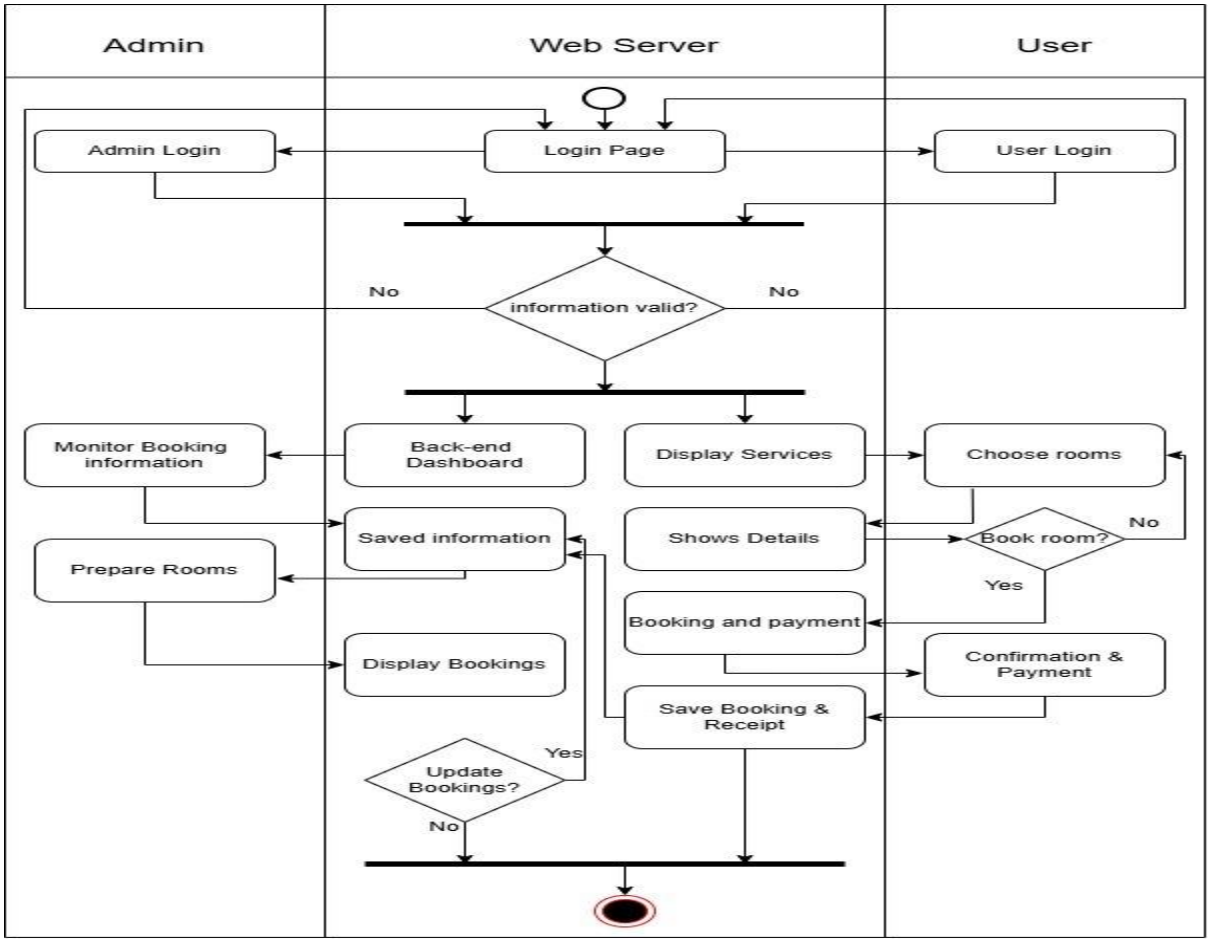
Use Case Diagram



ER Diagram



Class Diagram



Implemented Features

- Room booking and reservation system.
- Check-in and check-out management.
- Guest information and record storage.
- Automated billing and invoice generation.
- Real-time room availability status.
- User login and secure access system.
- Report generation for admin and managers.
- Payment tracking and history management.

Implementation

Technologies Used

- PHP and MYSQL for building the user interface
- AJAX for smooth, real-time updates without page reloads.

Frameworks and Tools

- HTML5, CSS3, JavaScript, Bootstrap for responsive and modern UI design.

Future Work

Possible Improvements

- Integrate online payment gateways for easier transactions.
- Add a mobile app version for better accessibility.
- Include advanced reporting and analytics for management decisions.
- Implement automated notifications via email or SMS for bookings and reminders.
- Add support for multiple languages to cater to international guests.

Next Development Steps

- Upgrade the database to handle larger amounts of data efficiently.
- Implement AI-based recommendation systems for personalized guest services.
- Introduce a loyalty program module for returning customers.
- Enhance system security with advanced authentication and encryption.
- Connect the system with third-party travel and booking platforms for wider reach.

Project Timeline

Phase	Activities	Timeline
1. Planning & Requirement Analysis	Define objectives, scope and requirements.	1–4 Nov
2. System Design	Prepare Use Case Diagram, ER Diagram and UI design	5–8 Nov
3. Database Development	Create MySQL database, tables, PK and FK relationships	9–12 Nov
4. Admin & User Module	Develop registration, login and dashboard.	13–16 Nov
5. Hotel & Room Management	Develop hotel, room and basic booking features.	17–19 Nov
6. Initial Testing & Review	Test core features and review project progress.	20 Nov

Conclusion

The Hotel Management System successfully automates key hotel operations such as room booking, check-in/check-out, billing, and guest record management. It reduces manual errors, saves time, and improves overall efficiency for hotel staff. Through this project, valuable insights were gained on system design, database management, and the importance of user-friendly interfaces in enhancing customer satisfaction.

References

- [1] M. S. M. Khan, S. Kashem, and M. M. Khan, "Development of a Comparison-Based Hotel and Resort Booking System in Bangladesh," *Journal of Software Engineering and Applications*, vol. 14, pp. 133–149, May 2024.
- [2] M. Ara, M. Foysal, M. A. Huda, and S. Bairagi, "A Model to Develop Hotel Management System to Optimize Revenue," *Khulna University Studies*, vol. 19, no. 2, pp. 167–188, Dec. 2025.
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